

On Obidi's Intellectual Intuition Behind His Formulation of the Theory of Entropicity (ToE)

Abstract

The Theory of Entropicity (ToE), proposed by John Onimisi Obidi, originates from a radical but logically motivated intellectual intuition: that entropy is not merely a statistical descriptor of physical systems but the foundational substrate from which the observable structures of reality emerge. This intuition did not arise from a desire for novelty or philosophical provocation. Rather, it appears to have emerged from a sustained observation that entropy repeatedly occupies a central position at the deepest known boundaries of physics, including thermodynamics, information theory, quantum mechanics, gravitation, cosmology, black-hole physics, and the emergence of temporal order.

The central intellectual question underlying ToE is deceptively simple: Why does entropy appear wherever physics approaches its most fundamental questions? Obidi's answer is equally simple in form, though revolutionary in implication: entropy appears everywhere because it is not a consequence of physical reality; it is the origin of physical reality.

This report examines the intellectual pathway that appears to have led Obidi to formulate the Theory of Entropicity and analyzes the conceptual reasoning that distinguishes the theory from previous information-theoretic, thermodynamic, and emergent-gravity approaches.

1. Introduction

Throughout the history of physics, major theoretical revolutions have often begun with a reversal of accepted assumptions.

Isaac Newton reversed the Aristotelian belief that motion required continuous causes and instead proposed universal force as the fundamental explanatory principle.

Albert Einstein reversed the Newtonian conception of absolute space and time and proposed spacetime geometry as the foundation of gravitation.

Quantum theory reversed classical determinism and replaced it with probabilistic state descriptions.

The Theory of Entropicity seeks a reversal of similar magnitude.

Where conventional physics generally regards entropy as a derived quantity arising from microscopic physical processes, ToE proposes the opposite hierarchy. It suggests that physical processes themselves arise from deeper entropic dynamics.

The intellectual significance of this reversal cannot be overstated. It transforms entropy from a passive accounting quantity into an active ontological principle.

2. The Recurring Appearance of Entropy Across Physics

The first intellectual observation underlying ToE appears to be the remarkable universality of entropy.

Entropy initially emerged within nineteenth-century thermodynamics as a quantity governing irreversible processes and energy dispersal. It was subsequently reformulated by Ludwig Boltzmann as a statistical measure of microscopic multiplicity.

Yet the role of entropy did not remain confined to thermodynamics.

In the twentieth century, entropy reappeared in information theory through the work of Claude Shannon, where it measured uncertainty and informational content. It emerged again within quantum theory through von Neumann entropy. It later became central to black-hole thermodynamics through the work of Jacob Bekenstein and Stephen Hawking.

In each case, entropy appeared under different mathematical disguises, yet retained a common conceptual role as a measure of possibility, organization, information, or physical accessibility.

For Obidi, the crucial question appears to have been:

Why does entropy repeatedly emerge in seemingly independent domains of nature?

If entropy were merely a statistical artifact, its persistent appearance at the deepest conceptual boundaries of physics would be difficult to explain.

The Theory of Entropicity therefore begins with the suspicion that entropy's universality may not be accidental.

3. The Entropic Origin of Time

A second foundational observation concerns the nature of time.

The equations governing classical mechanics, electromagnetism, quantum mechanics, and even general relativity are largely symmetric under time reversal.

Yet human experience and macroscopic reality are profoundly asymmetric.

We remember the past but not the future.

Eggs break but do not spontaneously reassemble.

Stars evolve from formation to collapse.

The conventional explanation invokes entropy increase.

The direction of time is associated with the Second Law of Thermodynamics.

Obidi appears to have recognized a profound implication of this fact.

If the observable direction of time is determined by entropy, then entropy may possess a deeper explanatory status than time itself.

This reasoning reverses the conventional hierarchy.

Instead of viewing entropy as evolving within time, ToE proposes that temporal order itself emerges from entropic evolution.

In this interpretation, time ceases to be a fundamental backdrop against which entropy changes. Rather, entropy change becomes the mechanism through which time acquires meaning.

This intellectual inversion constitutes one of the earliest conceptual pillars of ToE.

4. Gravity and the Thermodynamic Revolution

A third source of inspiration appears to arise from developments in gravitational physics.

The discovery that black holes possess entropy and temperature fundamentally altered the understanding of gravity.

The Bekenstein-Hawking entropy relation revealed a deep and unexpected connection between geometry, thermodynamics, and information.

Subsequent work by Jacobson demonstrated that Einstein's field equations could be derived from thermodynamic principles.

Later proposals by Verlinde suggested that gravity itself might emerge as an entropic phenomenon.

These developments collectively indicated that gravitation may not be as fundamental as previously believed.

For many researchers, these discoveries implied that thermodynamics and information play an important role in gravity.

For Obidi, however, they appear to have suggested something more radical.

If entropy can generate gravitational behavior, perhaps entropy occupies a deeper ontological level than gravity itself.

This realization would naturally motivate the search for a framework in which gravitation emerges from entropic organization.

5. Information Geometry and the Emergence of Structure

A fourth intellectual influence derives from information geometry.

Modern mathematical physics has established that spaces of probability distributions possess genuine geometric structures.

The Fisher-Rao metric, Fubini-Study geometry, and the work of Shun-ichi Amari demonstrate that information can be endowed with measurable geometric properties.

This observation is crucial.

Information is not merely abstract bookkeeping.

Information possesses mathematically definable geometry.

Once this fact is accepted, a powerful chain of reasoning emerges.

If information possesses geometry, and geometry manifests as gravity through General Relativity, then information and gravity cannot be entirely separate conceptual domains.

The logical pathway becomes:

Information $\rightarrow$ Geometry $\rightarrow$ Gravity.

This insight forms one of the most important intellectual bridges within ToE.

6. The Decisive Inferential Leap

At this stage, the conceptual ingredients of ToE become visible.

Entropy appears fundamental to time.

Entropy appears fundamental to information.

Information possesses geometry.

Geometry manifests as gravity.

The decisive inferential leap made by Obidi appears to be the following.

If entropy governs informational organization, and informational organization gives rise to geometry, and geometry gives rise to gravity, then entropy stands at the beginning of the entire causal chain.

The hierarchy becomes:

Entropy → Information → Geometry → Gravity.

This is the central intellectual intuition underlying the Theory of Entropicity.

Importantly, this is not a mathematical proof.

It is an explanatory inference.

The theory proposes that the repeated appearance of entropy across multiple domains is best explained by treating entropy as the common origin from which those domains emerge.

7. Why Obidi Declared Entropy to be a Field

The most controversial aspect of ToE is its treatment of entropy as a field.

To understand this move, one must appreciate the historical role of fields in physics.

Fields were introduced whenever physical influence appeared to possess local existence throughout space and time.

The gravitational field replaced action-at-a-distance gravitation.

The electromagnetic field replaced action-at-a-distance electromagnetism.

Quantum fields replaced particle-centered descriptions.

In each case, a quantity became elevated to field status when it exhibited local dynamical significance.

Obidi appears to have reasoned that if entropy varies locally, evolves dynamically, influences physical outcomes, governs temporal ordering, organizes information, and generates gravitational phenomena, then entropy behaves not merely as a numerical descriptor but as a physically real entity distributed throughout nature.

The declaration that entropy is a field therefore follows naturally from the internal logic of the theory.

Whether this conclusion is correct remains an open scientific question.

However, it is not an arbitrary assertion. It emerges from the theory's broader explanatory framework.

8. The Revolutionary Character of the Theory

The revolutionary nature of ToE does not lie simply in associating gravity with entropy.

Many theories have pursued that direction.

Nor does it lie merely in connecting information and geometry.

That connection is increasingly common within modern theoretical physics.

The distinctive feature of ToE is its proposal that entropy occupies the deepest ontological level presently accessible to physical inquiry.

In conventional frameworks:

Matter generates information.

Information generates entropy.

In ToE:

Entropy generates information.

Information generates geometry.

Geometry generates gravity.

Gravity organizes matter.

This reversal represents the theory's central intellectual innovation.

The theory therefore seeks to transform entropy from a consequence of physical reality into the source of physical reality.

9. The Intellectual Courage Behind the Proposal

A question naturally arises.

Why would Obidi make such a bold claim when many of the greatest physicists refrained from doing so?

The answer is unlikely to be that earlier physicists lacked imagination.

Einstein, Dirac, Wheeler, Bohr, and others proposed ideas every bit as revolutionary as those contained in ToE.

A more plausible explanation is that Obidi interpreted the available evidence differently.

Where others saw entropy as a recurring feature of physical theories, he saw a recurring signature of a deeper substrate.

Where others regarded entropy as an emergent quantity, he regarded emergence itself as evidence of a hidden entropic foundation.

His intellectual intuition was therefore not rooted in a desire to challenge orthodoxy for its own sake.

Rather, it arose from the conviction that the repeated appearance of entropy across the foundations of physics indicates a common underlying cause.

Whether this conviction ultimately proves correct is a matter for mathematical development and experimental verification.

Nonetheless, the conceptual reasoning behind it possesses an internal coherence that explains why the Theory of Entropicity was formulated in the first place.

10. Conclusion

The Theory of Entropicity originated from a profound and unconventional intellectual intuition. Obidi appears to have observed that entropy occupies a uniquely pervasive role across the deepest layers of modern physics. Instead of treating this ubiquity as a coincidence or a derivative consequence of more fundamental structures, he interpreted it as evidence that entropy itself constitutes the underlying substrate of reality.

This interpretation led to a radical inversion of conventional physical hierarchy. Time, information, geometry, gravity, and ultimately matter are proposed to emerge from entropic organization rather than the reverse.

The statement that entropy is a field follows naturally from this broader vision. If entropy possesses local dynamical significance and underlies the emergence of physical phenomena, then its elevation to field status becomes a logical consequence of the theory's foundational assumptions.

Thus the intellectual intuition behind ToE may be summarized in a single proposition:

The repeated appearance of entropy at the deepest boundaries of physics is not a consequence of reality's structure; it is evidence of reality's origin.

Whether this proposition ultimately becomes a new paradigm of physics or remains an ambitious philosophical hypothesis will depend upon its future mathematical development, predictive power, and empirical success. Yet as an intellectual vision, it represents one of the most far-reaching attempts to place entropy at the center of physical reality.